MIL-B-49430/3D(ER)

15 June 1987

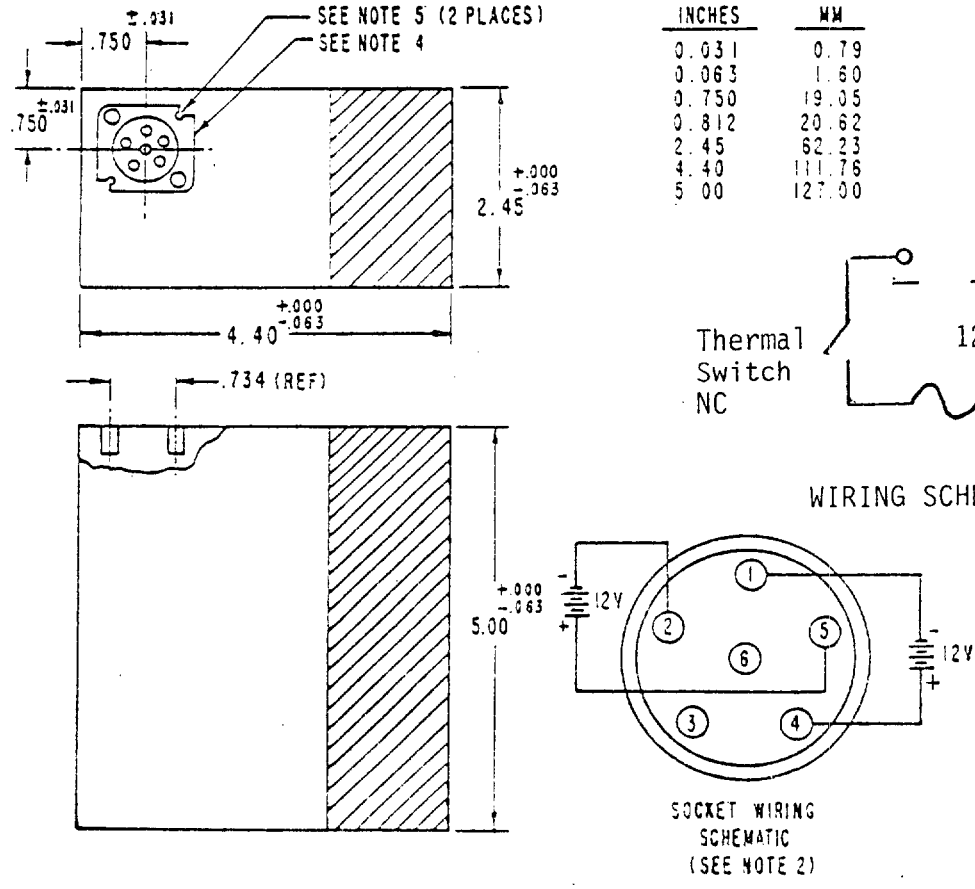
SUPERSEDING
MIL-B-49430/3C(ER)
4 January 1985

MILITARY SPECIFICATION SHEET

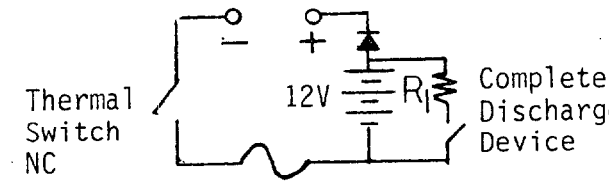
BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5590/U

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the battery described herein shall consist of this specification and the latest issue of MIL-B-49430(ER).



INCHES	MM
0.031	0.79
0.063	1.60
0.750	19.05
0.812	20.62
2.45	62.23
4.40	111.76
5.00	127.00



WIRING SCHEMATIC FOR EACH LEG

Figure 1. Battery dimensions and wiring schematic.

FSC 6135

AMSC N/A

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NOTES:

1. All dimensions are in inches.

2. Socket wiring schematic to be printed on jacket in suitable location near the socket on the terminal face of the battery.

3. Top of the connector socket is to be mounted flush with the top surface of the battery case.

4. The battery terminal shall consist of a miniature circular battery connector per Dwg. SC-C-179492. Connector shall be mounted in accordance with referenced drawing.

5. Connector socket body configuration shall accept two (2) each 0.156 inch diameter guidepost as shown. Depth of cavity shall be 0.622 inch.

6. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.

7. R_1 value shall be between 140 ohms and 200 ohms and have a power rating of 2.0 watts.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 9.15 volts.

Minimum voltage for capacity: 6.00 volts.

The battery shall contain a 2.25 ampere non-replaceable slow blow fuse in each negative leg.

A diode shall be wired in series with the cells in each individual 12 volt section to prevent charging or flow of current into the battery. See paragraph 3.5.5.2 of the Basic Specification

Terminals: Six (6) hole socket type with no obstruction of any holes (See fig. 1).

Weight (maximum): 2.25 pounds (1021 grams).

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Capacity Tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the maximum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in hours to 20.00 volts
I	3.2
L	1.8
H	5.6
HT	5.4
LT	1.6
T	3.0
D	3.0
ILR	1000

Initial Voltage delay: During the I, H, HT, T, D, and ILR discharge tests of the two (2) 12 volt sections connected in series under the designated loads, initial voltage below the 20.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 20.00 volts cannot exceed a 60 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery on each leg as described in paragraphs 3.5.5 and 3.5.5.1 of the Basic Specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para. 3.5.5.3 of the basic specification.

Connector: The connector shall be in accordance with SC-C-179492 with the following exception: The durability test described in paragraph 4h. shall be performed 75 times with connector SC-D-882165, 75 times with connector SC-C-179493, and 100 times with SM-D-687888.

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METHOD OF EXAMINATION AND TESTS:

Capacity Tests:

(1) Storage: Details on storage conditions for all specified capacity tests are described in basic specification.(See 4.8.9)

(2) Discharge:

Cells: Discharge at 2.0 amps constant current at 70±5°F for a minimum of 3.0 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 6.0 hours into voltage reversal.

Batteries: For all capacity tests except the H, HT and ILR, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I capacity test the batteries shall be force discharged at one ampere constant current for 6.4 hours immediately after zero volts is obtained. There shall be no venting, fire or explosion when the batteries are force discharged. For the ILR test a fixed resistance of 4700 ohms shall be used to a cut-off voltage of 20.00 volts or 1000 hours.

Closed circuit voltage: Load resistance of 14.0 ohms shall be used between socket holes 1 and 4, and 2 and 5. Voltage shall be 10.00 volts or greater within five seconds.

Custodian:
Army - ER

Preparing Activity:
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(Project 6135-A201-03)

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